

Solving 3D exterior Helmholtz problems in unbounded domains via VEM/BEM coupling

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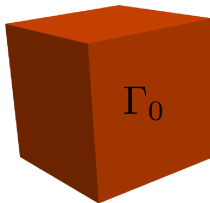
Matera, 20 – 24 July, 2026



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The model problem - 1

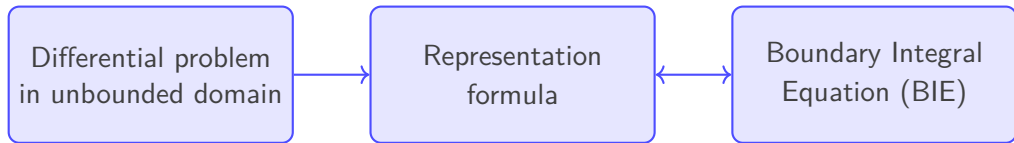
Let Ω_0 be a bounded polyhedral domain in $\mathbb{R}^3$. We consider the following exterior Dirichlet problem:



$$\begin{cases} -\Delta u_s - \kappa^2 u_s = f, & \text{in } \mathbb{R}^3 \setminus \overline{\Omega}_0, \\ u_s = 0, & \text{on } \Gamma_0 := \partial\Omega_0, \\ \lim_{r \rightarrow \infty} r(\partial_r - i\kappa)u_s = 0, & \end{cases} \quad (1)$$

where u_s is the (unknown) scattered field, $\kappa \in \mathbb{R}_+$ is the wavenumber, $f \in L^2_{\text{loc}}(\mathbb{R}^3 \setminus \overline{\Omega}_0)$ is a given function, and $r = \|\mathbf{x}\|$ with $\mathbf{x} = (x_1, x_2, x_3)^T$.

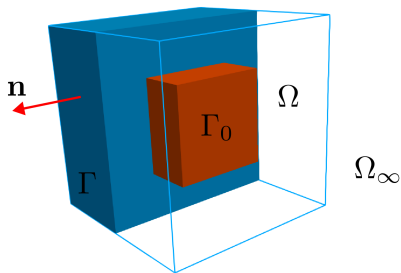
BEM workflow



The representation formula is computationally expensive: evaluating u_s at any $\mathbf{x} \in \mathbb{R}^3 \setminus \overline{\Omega}_0$ requires computing boundary integrals over Γ_0 .

The model problem - 2

We introduce an artificial boundary Γ which allows to decompose $\mathbb{R}^3 \setminus \overline{\Omega}_0$ into a finite computational domain Ω and an infinite residual domain Ω_∞ .



$$\left\{ \begin{array}{ll} -\Delta u - \kappa^2 u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \Gamma_0, \\ \gamma^- u = \gamma^+ u_\infty, & \text{on } \Gamma, \\ \partial_{\mathbf{n}}^- u = \partial_{\mathbf{n}}^+ u_\infty, & \text{on } \Gamma, \\ -\Delta u_\infty - \kappa^2 u_\infty = 0, & \text{in } \Omega_\infty, \\ \lim_{r \rightarrow \infty} r(\partial_r - i\kappa)u_\infty = 0. & \end{array} \right.$$

where u is the restriction of u_s to Ω , u_∞ is the restriction of u_s to Ω_∞ , $\gamma^\pm$ is the exterior/interior trace operator and $\partial_{\mathbf{n}}^\pm := \mathbf{n} \cdot \gamma^\pm \nabla$.

VEM/BEM coupling

VEM Suited for problems in bounded domains with complex geometries.

BEM Transforms a differential problem in an unbounded domain into an integral equation on the boundary.

Why don't we take the best from both worlds?

Kirchhoff's representation formula

It is well-known that u_∞ admits the following integral representation:

$$u_\infty(\mathbf{x}) = \int_\Gamma \frac{\partial G_\kappa}{\partial \mathbf{n}_y}(\mathbf{x}, \mathbf{y}) \psi(\mathbf{y}) d\mathbf{y} - \int_\Gamma G_\kappa(\mathbf{x}, \mathbf{y}) \lambda(\mathbf{y}) d\mathbf{y}, \quad \mathbf{x} \in \Omega_\infty$$

where

$$G_\kappa(\mathbf{x}, \mathbf{y}) = \frac{e^{i\kappa\|\mathbf{x}-\mathbf{y}\|}}{4\pi\|\mathbf{x}-\mathbf{y}\|}$$

and

$$\psi := \gamma^- u = \gamma^+ u_\infty, \quad \lambda := \partial_{\mathbf{n}}^- u = \partial_{\mathbf{n}}^+ u_\infty \quad \text{on } \Gamma.$$

Boundary Integral Operators

We introduce, for every $\mathbf{x} \in \Gamma$

$$V_\kappa: H^{-1/2}(\Gamma) \rightarrow H^{1/2}(\Gamma), \quad V_\kappa \lambda(\mathbf{x}) := \int_\Gamma G_\kappa(\mathbf{x}, \mathbf{y}) \lambda(\mathbf{y}) \, d\mathbf{y},$$

$$K_\kappa: H^{1/2}(\Gamma) \rightarrow H^{1/2}(\Gamma), \quad K_\kappa \psi(\mathbf{x}) := \int_\Gamma \frac{\partial G_\kappa}{\partial \mathbf{n}_\mathbf{y}}(\mathbf{x}, \mathbf{y}) \psi(\mathbf{y}) \, d\mathbf{y},$$

$$K'_\kappa: H^{-1/2}(\Gamma) \rightarrow H^{-1/2}(\Gamma), \quad K'_\kappa \lambda(\mathbf{x}) := \int_\Gamma \frac{\partial G_\kappa}{\partial \mathbf{n}_\mathbf{x}}(\mathbf{x}, \mathbf{y}) \lambda(\mathbf{y}) \, d\mathbf{y},$$

$$W_\kappa: H^{1/2}(\Gamma) \rightarrow H^{-1/2}(\Gamma), \quad W_\kappa \psi(\mathbf{x}) := -\frac{\partial}{\partial \mathbf{n}_\mathbf{x}} \int_\Gamma \frac{\partial G_\kappa}{\partial \mathbf{n}_\mathbf{y}}(\mathbf{x}, \mathbf{y}) \psi(\mathbf{y}) \, d\mathbf{y}.$$

Boundary Integral Equations

By applying the traces operators to the integral representation formula, we get the following two boundary equations:

$$0 = \left(\frac{1}{2}I - K_{\kappa} \right) \psi + V_{\kappa} \lambda, \quad \text{on } \Gamma, \quad (2)$$

$$\lambda = -W_{\kappa} \psi + \left(\frac{1}{2}I - K'_{\kappa} \right) \lambda, \quad \text{on } \Gamma. \quad (3)$$

We use (2)-(3) to restrict the original problem to the bounded region Ω . The new problem now is:

The model problem - 3

We seek u such that

$$\left\{ \begin{array}{ll} -\Delta u - \kappa^2 u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \Gamma_0, \\ \gamma^- u = \gamma^+ u_\infty, & \text{on } \Gamma, \\ \partial_{\mathbf{n}}^- u = \partial_{\mathbf{n}}^+ u_\infty, & \text{on } \Gamma, \\ \left(\frac{1}{2}\mathbf{I} - \mathbf{K}_\kappa\right) \psi + \mathbf{V}_\kappa \lambda = 0, & \text{on } \Gamma, \\ -\mathbf{W}_\kappa \psi + \left(\frac{1}{2}\mathbf{I} - \mathbf{K}'_\kappa\right) \lambda = \lambda, & \text{on } \Gamma. \end{array} \right.$$

Variational formulation - 1

Define $H_{0,\Gamma_0}^1(\Omega) = \{v \in H^1(\Omega) : v|_{\Gamma_0} = 0\}$ and $\mathbb{X} := H_{0,\Gamma_0}^1(\Omega) \times H_0^{1/2}(\Gamma) \times H^{-1/2}(\Gamma)$.

We look for $(u, \psi, \lambda) \in \mathbb{X}$ such that

$$a(u, v) - \kappa^2 m(u, v) - \langle \lambda, \gamma v - \varphi \rangle - \langle \lambda, \varphi \rangle + \langle \mu, \gamma u - \psi \rangle = m(f, v)$$

is satisfied for all $(v, \varphi, \mu) \in \mathbb{X}$, where

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v, \quad m(u, v) := \int_{\Omega} u v, \quad \langle \cdot, \cdot \rangle_{\Gamma} := \text{Duality product } H^{-1/2}(\Gamma) - H^{1/2}(\Gamma).$$

Employ the weak form of the boundary equations (2)-(3) into the **fourth term** of the left hand side.

Variational formulation - 2

Find $(u, \psi, \lambda) \in \mathbb{X}$ such that

$$\mathcal{A}_\kappa((u, \psi, \lambda), (v, \varphi, \mu)) = m(f, v), \quad (4)$$

for all $(v, \varphi, \mu) \in \mathbb{X}$,

where

$$\begin{aligned} \mathcal{A}_\kappa((u, \psi, \lambda), (v, \varphi, \mu)) &:= \mathbf{A}_\kappa((u, \psi, \lambda), (v, \varphi, \mu)) + \langle \mathbf{W}_\kappa \psi, \varphi \rangle_\Gamma \\ &\quad + \langle \mu, \mathbf{V}_\kappa \lambda \rangle_\Gamma - \langle \lambda, (\tfrac{1}{2}\mathbf{I} - \mathbf{K}_\kappa) \varphi \rangle_\Gamma + \langle \mu, (\tfrac{1}{2}\mathbf{I} - \mathbf{K}_\kappa) \psi \rangle_\Gamma, \end{aligned}$$

and

$$\mathbf{A}_\kappa((u, \psi, \lambda), (v, \varphi, \mu)) := a(u, v) - \kappa^2 m(u, v) - \langle \lambda, \gamma v - \varphi \rangle_\Gamma + \langle \mu, \gamma u - \psi \rangle_\Gamma.$$

Well-posedness of the continuous problem

Theorem

Assume that κ^2 is not an eigenvalue of the Laplacian in $\Omega \cup \overline{\Omega}_0$ with a Dirichlet boundary condition on Γ . Then, problem (4) is well posed ^a.

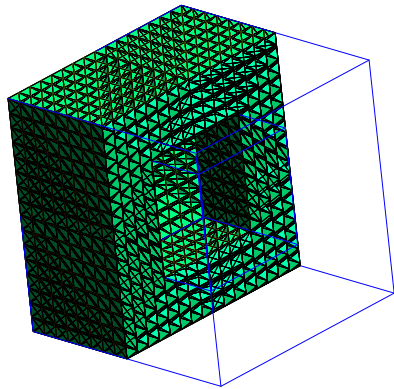
^aG. N. Gatica, S. Meddahi, *Coupling of virtual element and boundary element methods for the solution of acoustic scattering problems*, J. Numer. Math 28(2020), pp. 223-245.

Discrete setting

Let

- $\{\mathcal{T}_h\}$ be a family of polyhedral decompositions of $\bar{\Omega}$ such that $\bigcup_{E \in \mathcal{T}_h} \bar{E} = \bar{\Omega}$. Each element $E \in \mathcal{T}_h$ has diameter $h_E \leq h$;
- $\mathcal{F}_h$ the set of faces of $\mathcal{T}_h$ contained in Γ .

Moreover, we suppose that the mesh satisfies the usual assumptions required by the VEM^a.



^aB. Ahmad, A. Alsaedi, F. Brezzi, L.D. Marini, A. Russo, *Equivalent projectors for virtual element methods*, Comput. Math. Appl. 66 (2013), n. 3, 376–391

VEM spaces

We introduce the following local VEM spaces: for any $E \in \mathcal{T}_h$ and $F \subset \partial E$,

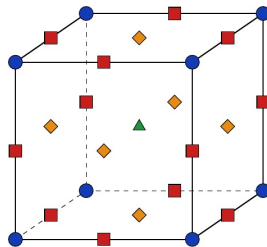
$$X_h^k(F) := \{v \in H^1(F) : v|_e \in \mathbb{P}_k(e), \forall e \subseteq \partial F, \Delta v \in \mathbb{P}_k(F), \Pi_k^{0,F} v - \Pi_k^{\nabla,F} v \in \mathbb{P}_{k-2}(F)\},$$

$$X_h^k(\partial E) := \{v \in C^0(\partial E) : v|_F \in X_h^k(F), \forall F \subseteq \partial E\},$$

$$W_h^k(E) := \{v \in H^1(E) : v|_{\partial E} \in X_h^k(\partial E), \Delta v \in \mathbb{P}_k(E), \Pi_k^{0,E} v - \Pi_k^{\nabla,E} v \in \mathbb{P}_{k-2}(E)\}.$$

The degrees of freedom of the space $W_h^k(E)$ are:

- the values at the vertices of E ,
- the moments of order $\leq k - 2$ on the edges of E ,
- the moments of order $\leq k - 2$ on the faces of E ,
- the moments of order $\leq k - 2$ on E .



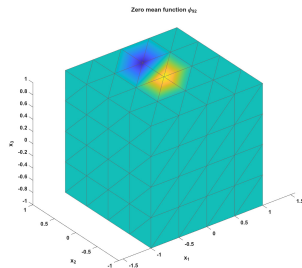
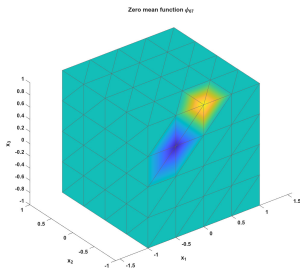
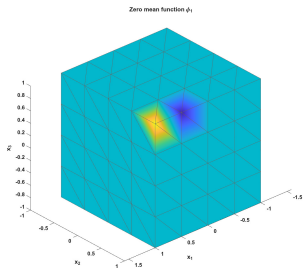
Dofs for $k = 2$.

Discrete boundary spaces

We now introduce the simplicial mesh $\mathfrak{F}_h$ of the interface Γ obtained by triangulating each polygon $F \in \mathcal{F}_h$.

$$\Lambda_h^{k-1} := \{\mu_h \in L^2(\Gamma) : \mu_h|_T \in \mathbb{P}_{k-1}(T), \forall T \in \mathfrak{F}_h\},$$

$$\Psi_h^k := \{\varphi_h \in C^0(\Gamma) : \varphi_h|_T \in \mathbb{P}_k(T), \forall T \in \mathfrak{F}_h\} \cap H_0^{1/2}(\Gamma).$$



A VEM/BEM scheme

Define the global VEM space $W_h^k := \{v \in H_{0,\Gamma_0}^1(\Omega) : v|_E \in W_h^k(E), \forall E \in \mathcal{T}_h\}$.

Find $(u_h, \psi_h, \lambda_h) \in \mathbb{X}_h := W_h^k \times \Psi_h^k \times \Lambda_h^{k-1}$ such that

$$\mathcal{A}_{h,\kappa}((u_h, \psi_h, \lambda_h), (v_h, \varphi_h, \mu_h)) = m_h(f, v_h) := \sum_{E \in \mathcal{T}_h} \int_E (\Pi_{k-1}^{0,E} f) (\Pi_{k-1}^{0,E} v_h) d\mathbf{x} \quad (5)$$

for all $(v_h, \varphi_h, \mu_h) \in \mathbb{X}_h$.

A VEM/BEM scheme

where

$$\begin{aligned} \mathcal{A}_{h,\kappa}((u_h, \psi_h, \lambda_h), (v_h, \varphi_h, \mu_h)) &:= \mathbf{A}_{h,\kappa}((u_h, \psi_h, \lambda_h), (v_h, \varphi_h, \mu_h)) \\ &\quad + \langle \mathbf{W}_\kappa \psi_h, \varphi_h \rangle_\Gamma + \langle \mu_h, \mathbf{V}_\kappa \lambda_h \rangle_\Gamma - \langle \lambda_h, (\tfrac{1}{2}\mathbf{I} - \mathbf{K}_\kappa) \varphi_h \rangle_\Gamma + \langle \mu_h, (\tfrac{1}{2}\mathbf{I} - \mathbf{K}_\kappa) \psi_h \rangle_\Gamma, \end{aligned}$$

and

$$\begin{aligned} \mathbf{A}_{h,\kappa}((u_h, \psi_h, \lambda_h), (v_h, \varphi_h, \mu_h)) &:= a_h(u_h, v_h) - \kappa^2 \sum_{E \in \mathcal{T}_h} m^E(\Pi_{k-1}^{0,E} u_h, \Pi_{k-1}^{0,E} v_h) \\ &\quad - \sum_{F \in \mathcal{F}_h} \langle \lambda_h, \Pi_{k-1}^{0,F}(\gamma v_h - \varphi_h) \rangle_\Gamma + \sum_{F \in \mathcal{F}_h} \langle \mu_h, \Pi_{k-1}^{0,F}(\gamma u_h - \psi_h) \rangle_\Gamma. \end{aligned}$$

The bilinear form a_h

$$a_h(v, z) := \sum_{E \in \mathcal{T}_h} a_h^E(v, z), \quad \forall v, z \in W_h^k,$$

$$a_h^E(v, z) := \int_E (\nabla \Pi_k^{\nabla, E} v) \cdot (\nabla \Pi_k^{\nabla, E} z) + S_h^E(v - \Pi_k^{\nabla, E} v, z - \Pi_k^{\nabla, E} z),$$

$$S_h^E(v, z) := \sum_{i=1}^{N_{\text{dof}}^E} \text{dof}_i(v) \text{dof}_i(z).$$

Order of convergence

Theorem

Assume that κ^2 is not an eigenvalue of the laplacian in $\Omega \cup \overline{\Omega}_0$ with a Dirichlet boundary condition on Γ . Let $u \in H_{0,\Gamma_0}^1(\Omega) \cap \prod_{E \in \mathcal{T}_h} H^{k+1}(E)$ and $f \in \prod_{E \in \mathcal{T}_h} H^k(E)$. Then there exist two constants $C > 0$, independent of h , and $h_0 > 0$ such that for any $h \leq h_0$ the following estimate holds ^a

$$\|(u, \psi, \lambda) - (u_h, \psi_h, \lambda_h)\|_{\mathbb{X}} \lesssim C h^k \sum_{E \in \mathcal{T}_h} \left(\|u\|_{H^{k+1}(E)} + \|f\|_{H^k(E)} \right).$$

^aG. N. Gatica, S. Meddahi, *Coupling of virtual element and boundary element methods for the solution of acoustic scattering problems*, J. Numer. Math 28(2020), pp. 223-245.

Insight on the implementation of the VEM/BEM coupling

We rewrite the system of equations as: find $(u_h, \psi_h, \lambda_h) \in \mathbb{X}_h$ such that

$$\begin{cases} a_h(u_h, v_h) - \kappa^2 m_h(u_h, v_h) - \langle \lambda_h, \gamma v_h \rangle_\Gamma = m_h(f, v_h), \\ \langle \mu_h, V_\kappa \lambda_h \rangle_\Gamma + \langle \mu_h, (-\frac{1}{2}\mathbf{I} - \mathbf{K}_\kappa) \psi_h \rangle_\Gamma + \langle \mu_h, \gamma u_h \rangle_\Gamma = 0, \\ \langle \mathbf{W}_\kappa \psi_h, \varphi_h \rangle_\Gamma + \langle (\frac{1}{2}\mathbf{I} + \mathbf{K}'_\kappa) \lambda_h, \varphi_h \rangle_\Gamma = 0, \end{cases}$$

for all $(v_h, \varphi_h, \mu_h) \in \mathbb{X}_h$. Moreover, we introduce

$$W_h^k = \text{span}\{\Phi_i\}, \quad \Psi_h^k = \text{span}\{\chi_i\}, \quad \Lambda_h^{k-1} = \text{span}\{\xi_i\}, \\ u_h(\mathbf{x}) = \sum \mathbf{u}_i \Phi_i(\mathbf{x}), \quad \psi_h(\mathbf{x}) = \sum \psi_i \chi_i(\mathbf{x}), \quad \lambda_h(\mathbf{x}) = \sum \lambda_i \xi_i(\mathbf{x}).$$

Global linear system

$$\begin{bmatrix} \mathbf{A}^{\Gamma\Gamma} - \kappa^2 \mathbf{M}^{\Gamma\Gamma} & \mathbf{A}^{\Gamma I} - \kappa^2 \mathbf{M}^{\Gamma I} & -\mathbf{Q} & \mathbf{O} \\ \mathbf{A}^{I\Gamma} - \kappa^2 \mathbf{M}^{I\Gamma} & \mathbf{A}^{II} - \kappa^2 \mathbf{M}^{II} & \mathbf{O} & \mathbf{O} \\ -\mathbf{Q}^T & \mathbf{O} & -\mathbf{V} & \frac{1}{2}\mathbf{N} + \mathbf{K} \\ \mathbf{O} & \mathbf{O} & \frac{1}{2}\mathbf{N}^T + \mathbf{K}^T & \mathbf{W} \end{bmatrix} \begin{bmatrix} \mathbf{u}^{\Gamma} \\ \mathbf{u}^I \\ \boldsymbol{\lambda} \\ \boldsymbol{\psi} \end{bmatrix} = \begin{bmatrix} \mathbf{f}^{\Gamma} \\ \mathbf{f}^I \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (6)$$

$$\mathbb{A}_{ij} = a_h(\Phi_j, \Phi_i)$$

Global linear system

$$\begin{bmatrix}
 \mathbb{A}^{\Gamma\Gamma} - \kappa^2 \mathbb{M}^{\Gamma\Gamma} & \mathbb{A}^{\Gamma I} - \kappa^2 \mathbb{M}^{\Gamma I} & -\mathbb{Q} & \mathbb{O} \\
 \mathbb{A}^{I\Gamma} - \kappa^2 \mathbb{M}^{I\Gamma} & \mathbb{A}^{II} - \kappa^2 \mathbb{M}^{II} & \mathbb{O} & \mathbb{O} \\
 -\mathbb{Q}^T & \mathbb{O} & -\mathbb{V} & \frac{1}{2}\mathbb{N} + \mathbb{K} \\
 \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbb{N}^T + \mathbb{K}^T & \mathbb{W}
 \end{bmatrix}
 \begin{bmatrix}
 \mathbf{u}^\Gamma \\
 \mathbf{u}^I \\
 \boldsymbol{\lambda} \\
 \boldsymbol{\psi}
 \end{bmatrix}
 =
 \begin{bmatrix}
 \mathbf{f}^\Gamma \\
 \mathbf{f}^I \\
 \mathbf{0} \\
 \mathbf{0}
 \end{bmatrix}
 \quad (6)$$

$$\mathbb{M}_{ij} = m_h(\Phi_j, \Phi_i)$$

Global linear system

$$\begin{bmatrix}
 \mathbb{A}^{\Gamma\Gamma} - \kappa^2 \mathbb{M}^{\Gamma\Gamma} & \mathbb{A}^{\Gamma I} - \kappa^2 \mathbb{M}^{\Gamma I} & -\mathbb{Q} & \mathbb{O} \\
 \mathbb{A}^{I\Gamma} - \kappa^2 \mathbb{M}^{I\Gamma} & \mathbb{A}^{II} - \kappa^2 \mathbb{M}^{II} & \mathbb{O} & \mathbb{O} \\
 -\mathbb{Q}^T & \mathbb{O} & -\mathbb{V} & \frac{1}{2}\mathbb{N} + \mathbb{K} \\
 \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbb{N}^T + \mathbb{K}^T & \mathbb{W}
 \end{bmatrix}
 \begin{bmatrix}
 \mathbf{u}^\Gamma \\
 \mathbf{u}^I \\
 \boldsymbol{\lambda} \\
 \boldsymbol{\psi}
 \end{bmatrix}
 =
 \begin{bmatrix}
 \mathbf{f}^\Gamma \\
 \mathbf{f}^I \\
 \mathbf{0} \\
 \mathbf{0}
 \end{bmatrix} \quad (6)$$

$$\mathbb{Q}_{ij} = \langle \xi_j, \Phi_i \rangle_\Gamma$$

Global linear system

$$\begin{bmatrix} \mathbb{A}^{\Gamma\Gamma} - \kappa^2 \mathbb{M}^{\Gamma\Gamma} & \mathbb{A}^{\Gamma I} - \kappa^2 \mathbb{M}^{\Gamma I} & -\mathbb{Q} & \mathbb{O} \\ \mathbb{A}^{I\Gamma} - \kappa^2 \mathbb{M}^{I\Gamma} & \mathbb{A}^{II} - \kappa^2 \mathbb{M}^{II} & \mathbb{O} & \mathbb{O} \\ -\mathbb{Q}^T & \mathbb{O} & -\mathbb{V} & \frac{1}{2}\mathbb{N} + \mathbb{K} \\ \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbb{N}^T + \mathbb{K}^T & \mathbb{W} \end{bmatrix} \begin{bmatrix} \mathbf{u}^\Gamma \\ \mathbf{u}^I \\ \boldsymbol{\lambda} \\ \boldsymbol{\psi} \end{bmatrix} = \begin{bmatrix} \mathbf{f}^\Gamma \\ \mathbf{f}^I \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (6)$$

$$\mathbb{V}_{ij} = \int_{\Gamma} \left(\int_{\Gamma} G_{\kappa}(\mathbf{x}, \mathbf{y}) \xi_j(\mathbf{y}) d\mathbf{y} \right) \xi_i(\mathbf{x}) d\mathbf{x}$$

Global linear system

$$\begin{bmatrix}
 \mathbb{A}^{\Gamma\Gamma} - \kappa^2 \mathbb{M}^{\Gamma\Gamma} & \mathbb{A}^{\Gamma I} - \kappa^2 \mathbb{M}^{\Gamma I} & -\mathbb{Q} & \mathbb{O} \\
 \mathbb{A}^{I\Gamma} - \kappa^2 \mathbb{M}^{I\Gamma} & \mathbb{A}^{II} - \kappa^2 \mathbb{M}^{II} & \mathbb{O} & \mathbb{O} \\
 -\mathbb{Q}^T & \mathbb{O} & -\mathbb{V} & \frac{1}{2}\mathbb{N} + \mathbb{K} \\
 \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbb{N}^T + \mathbb{K}^T & \mathbb{W}
 \end{bmatrix}
 \begin{bmatrix}
 \mathbf{u}^\Gamma \\
 \mathbf{u}^I \\
 \boldsymbol{\lambda} \\
 \boldsymbol{\psi}
 \end{bmatrix}
 =
 \begin{bmatrix}
 \mathbf{f}^\Gamma \\
 \mathbf{f}^I \\
 \mathbf{0} \\
 \mathbf{0}
 \end{bmatrix}
 \quad (6)$$

$$\mathbb{W}_{ij} = - \int_{\Gamma} \left(\frac{\partial}{\partial \mathbf{n}_{\mathbf{x}}} \int_{\Gamma} \frac{\partial G_{\kappa}}{\partial \mathbf{n}_{\mathbf{y}}}(\mathbf{x}, \mathbf{y}) \chi_j(\mathbf{y}) d\mathbf{y} \right) \chi_i(\mathbf{x}) d\mathbf{x}$$

Global linear system

$$\begin{bmatrix} \mathbb{A}^{\Gamma\Gamma} - \kappa^2 \mathbb{M}^{\Gamma\Gamma} & \mathbb{A}^{\Gamma I} - \kappa^2 \mathbb{M}^{\Gamma I} & -\mathbb{Q} & \mathbb{O} \\ \mathbb{A}^{I\Gamma} - \kappa^2 \mathbb{M}^{I\Gamma} & \mathbb{A}^{II} - \kappa^2 \mathbb{M}^{II} & \mathbb{O} & \mathbb{O} \\ -\mathbb{Q}^T & \mathbb{O} & -\mathbb{V} & \frac{1}{2}\mathbb{N} + \mathbb{K} \\ \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbb{N}^T + \mathbb{K}^T & \mathbb{W} \end{bmatrix} \begin{bmatrix} \mathbf{u}^\Gamma \\ \mathbf{u}^I \\ \boldsymbol{\lambda} \\ \boldsymbol{\psi} \end{bmatrix} = \begin{bmatrix} \mathbf{f}^\Gamma \\ \mathbf{f}^I \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (6)$$

$$\mathbb{K}_{ij} = \int_{\Gamma} \left(\int_{\Gamma} \frac{\partial G_{\kappa}}{\partial \mathbf{n}_{\mathbf{y}}}(\mathbf{x}, \mathbf{y}) \chi_j(\mathbf{y}) d\mathbf{y} \right) \xi_i(\mathbf{x}) d\mathbf{x}, \quad \mathbb{N}_{ij} = \int_{\Gamma} \chi_j(\mathbf{x}) \xi_i(\mathbf{x}) d\mathbf{x}$$

Global linear system

$$\begin{bmatrix}
 \mathbf{A}^{\Gamma\Gamma} - \kappa^2 \mathbf{M}^{\Gamma\Gamma} & \mathbf{A}^{\Gamma I} - \kappa^2 \mathbf{M}^{\Gamma I} & -\mathbf{Q} & \mathbb{O} \\
 \mathbf{A}^{I\Gamma} - \kappa^2 \mathbf{M}^{I\Gamma} & \mathbf{A}^{II} - \kappa^2 \mathbf{M}^{II} & \mathbb{O} & \mathbb{O} \\
 -\mathbf{Q}^T & \mathbb{O} & -\mathbf{V} & \frac{1}{2}\mathbf{N} + \mathbf{K} \\
 \mathbb{O} & \mathbb{O} & \frac{1}{2}\mathbf{N}^T + \mathbf{K}^T & \mathbf{W}
 \end{bmatrix}
 \begin{bmatrix}
 \mathbf{u}^{\Gamma} \\
 \mathbf{u}^I \\
 \boldsymbol{\lambda} \\
 \boldsymbol{\psi}
 \end{bmatrix}
 =
 \begin{bmatrix}
 \mathbf{f}^{\Gamma} \\
 \mathbf{f}^I \\
 \mathbf{0} \\
 \mathbf{0}
 \end{bmatrix} \quad (6)$$

$\mathbb{O}$ = matrix with all zero elements

A numerical test

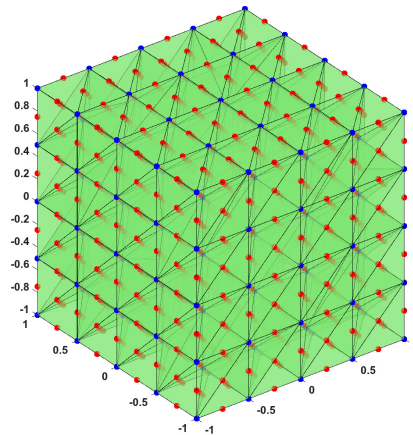
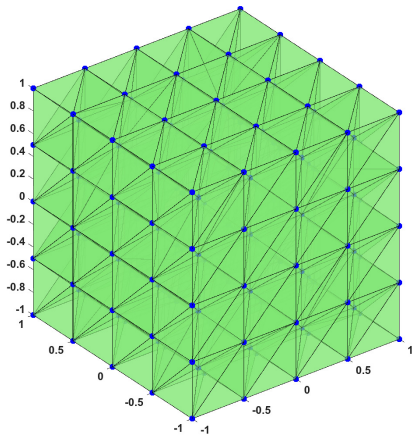
In our numerical tests, we consider the problem:

$$\begin{cases} -\Delta u_s - \kappa^2 u_s = 0, & \text{in } \mathbb{R}^3 \setminus \overline{\Omega}_0 \\ u_s = g, & \text{on } \Gamma_0, \\ \lim_{r \rightarrow \infty} r(\partial_r - i\kappa)u_s = 0, \end{cases} \quad (7)$$

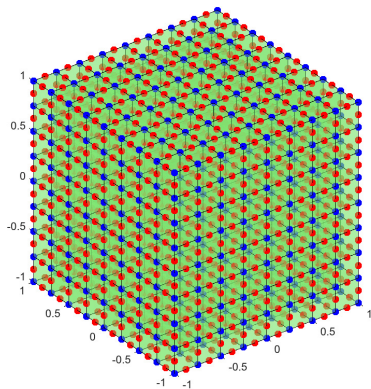
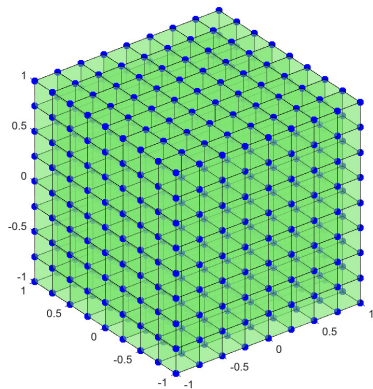
Let $\Omega_0 = [-0.5, 0.5]^3$ and $\Gamma = \partial([-1, 1]^3)$. We choose the Dirichlet datum g on Γ_0 such that the exact solution of problem (7) is given by

$$u(\mathbf{x}) = \frac{e^{i\kappa\|\mathbf{x}\|}}{\|\mathbf{x}\|^2} \left(1 + \frac{i}{\kappa\|\mathbf{x}\|} \right) x_3, \quad \text{for } \mathbf{x} \in \Omega.$$

Tetrahedral mesh for degree $k = 1$ and $k = 2$



Hexahedral mesh for degree $k = 1$ and $k = 2$



Convergence results, $\kappa = 1$

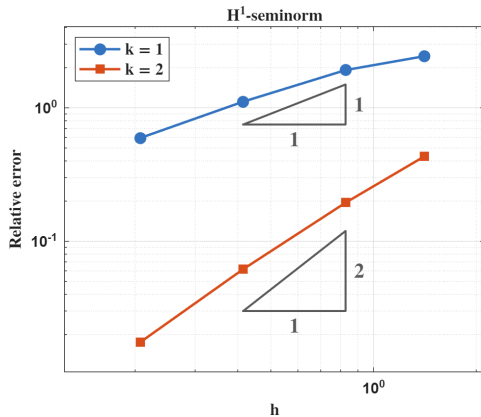
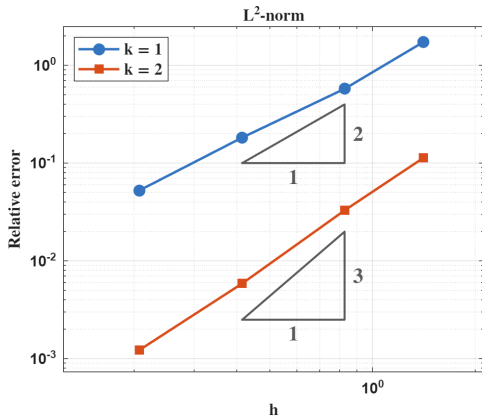


Figure: Tetrahedral mesh.

Convergence results, $\kappa = 1$

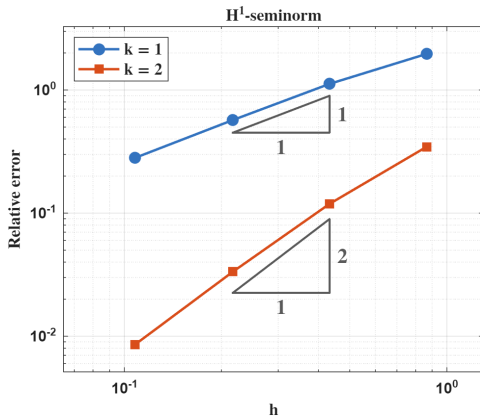
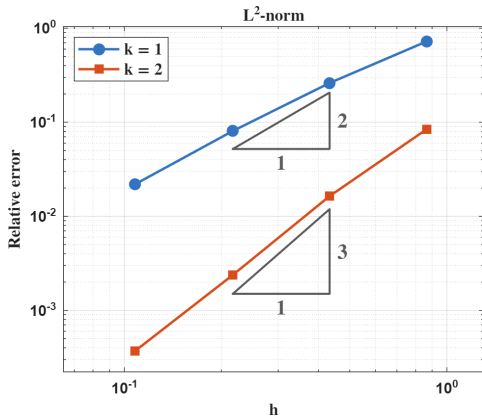


Figure: Hexahedral mesh.

Behavior of the exact and approximate solutions

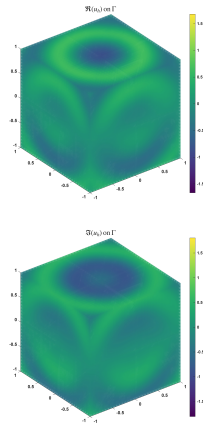
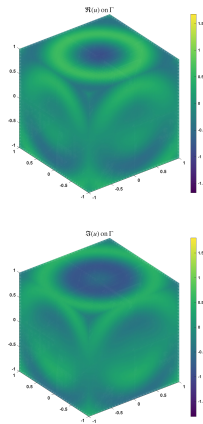
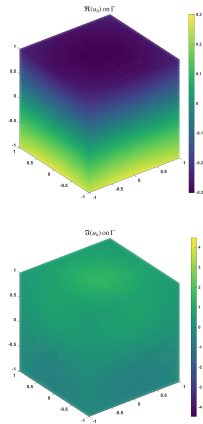
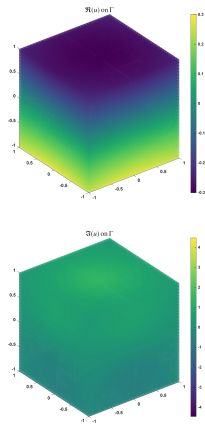


Figure: $\kappa = 1$

Figure: $\kappa = 10$